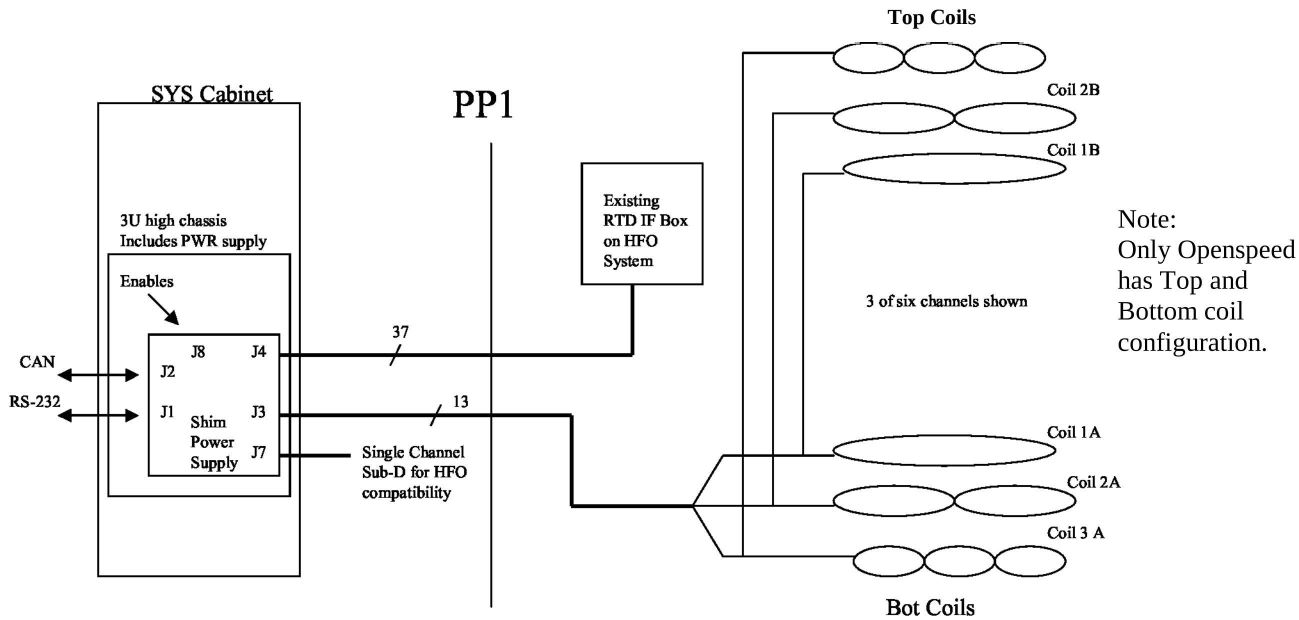
High Order Shim Supply Theory Of Operation

PURPOSE

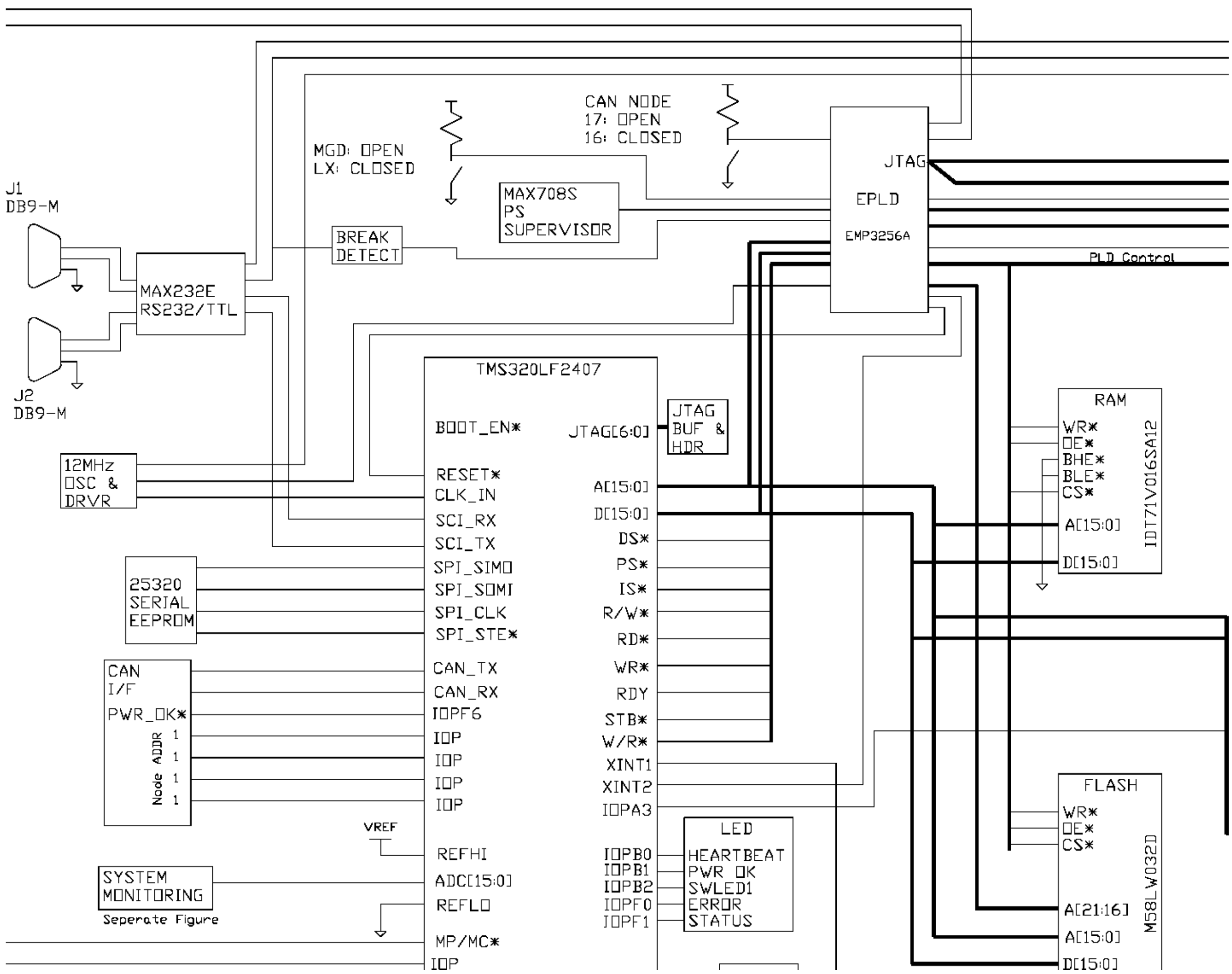
This Theory document details the new shim power supply used in the Open Speed, Twin, and 3T MRI imaging systems. This assembly:

- Provides 6, 4 amp shim channels for supplying current to coils at the magnet (Total absolute value of coil currents not to exceed 12 amps).
- Provisions for installing a smaller power supply for additional cost savings while meeting the requirements for HFO only systems.
- Provides an RS-232 interface for communication to the host when installed in a LX system.
- Provides a CAN (Controller Area Network) interface for communication when installed in a MGD system.
- Monitors 8, 4 wire RTD sensors to eliminate the Fluke temperature monitor used in Open Speed systems.
- Provides a high degree of system monitoring, and diagnostic capability.



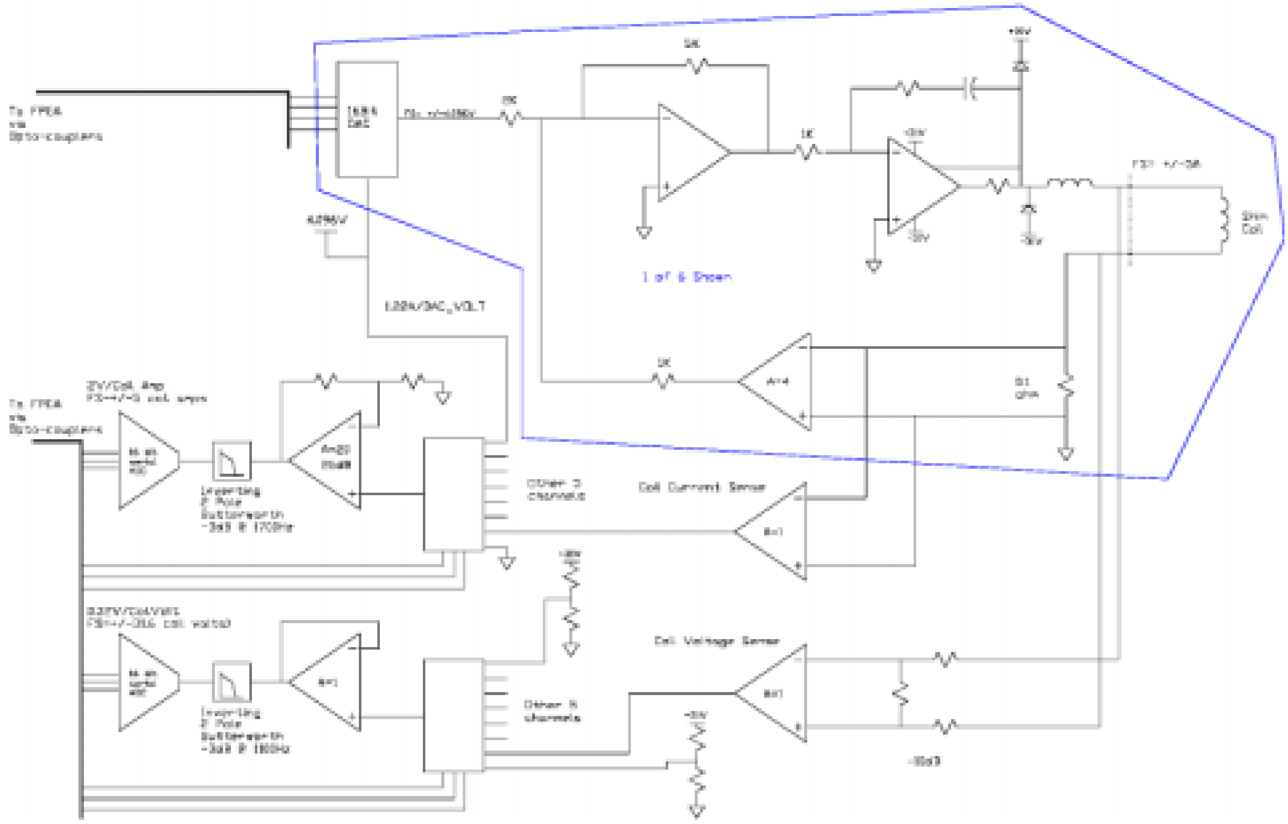
Top Level Block Diagram

BOARD BLOCK DIAGRAM

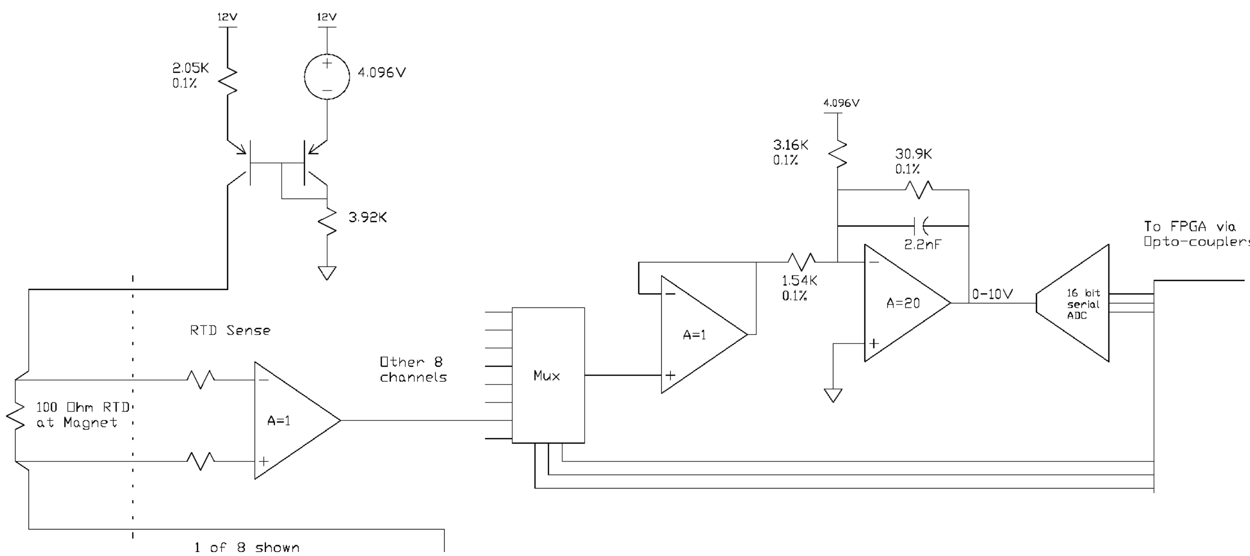


Board Block Diagram

COIL DRIVE ELECTRONICS



RTD TEMPERATURE MONITOR



RTD Temperature Electronics

OVERVIEW

Note that the connector designators indicated here, are for the chassis not the board.

POWER AND PACKAGING

The shim supply electronics will be housed in a 3U high chassis. The chassis will be powered from 120 volts AC via an IEC-320 receptacle on the rear of the chassis. The power supply will deliver +5volts, +/-15 volts, and +/31 volts to the board. The +/-15 and +/-31 volt supplies will be isolated from ground. Onboard regulators will generate other voltages required by the board.

Much of the digital electronics is a cut and paste from the MDS-CAN Translator Board, part number 2281054. A TMS320LF2407A (DSP from TI) will control all board functions. This device will be referred to as the uC (microcontroller) in the rest of this document.

A serial EEPROM will be used to store configuration/calibration information during power cycling.

COMMUNICATION

The board will have both an RS-232 and CAN (Controller Area Network) interface. A switch on the front of the chassis will select between LX and MGD systems. This switch will be labeled with MGD/LX. There are two RS232 ports, J1 and J2). J1 is connected to a UART chip (via the RS-232 level translator) while J2 is connected to the serial port on the uC (also through the level translator). J2 will be a debug/monitor port regardless of the boot mode.

A command set similar to that used on existing LX systems will be supported over the RS-232 interface to make this new shim supply backward compatible as a replacement for the RRI (Resonant Research Inc.) shim supply. Downloading executable files (images) via RS-232 from the host will be supported on LX systems, as well as over the CAN on MGD systems. A software patch will be required on older systems when upgrading to the new shim supply.

A break detector will monitor the RS-232 receive channel (from the host) to allow the host to reset the shim supply.

COIL DRIVE

The shim coil current drive section consists of 6 power op-amp’s, each capable of driving 4 amps through the shim coil connected to it. The absolute maximum current from all the outputs must not exceed 12 amps. A separate DAC exists for each of the 6 output amplifiers. Shim coil voltage and current is monitored by the uC. This information is used to subtract offset and gain errors in the power electronics, and for calculating coil resistance to verify coil integrity. The maximum power dissipation from all power amplifiers combined, under normal operating conditions, will be approximately 200 watts. The power amplifiers will be mounted on a forced air heatsink to remove the heat generated. The heat sink temperature will be monitored and the supply voltage to the power amplifiers will be interrupted if the temperature exceeds 66 degrees C. The shim coils will connect to the rear of the chassis via a 37 pin circular connector (J3) similar to that used on existing Twin/3T (2257702) supply. Additionally, a single channel (Duplicate of channel 5) will also be brought out through a 9 pin female DB connector (J7) on the rear of the chassis to provide compatibility with existing HFO Z2 supplies.

Heat sink temperature is monitored by the uC, if excessive temperature is detected then DAC outputs will be zeroed and/or the power supplied to the power amplifiers removed. The severity of the action depends on the degree of over temperature, for example an information message can be logged if an OT1 (over temp 1) condition exists, but exceeding OT2 should result in the supply voltage being removed from the output amplifiers. It is assumed that the second, higher temperature threshold (OT2) would never be reached unless the coil drive hardware malfunctioned, providing the uC was alive and in control. To maintain uC control, a 2 second watchdog timer is implemented in the FPGA to turn off the rail voltage if the uC is not periodically resetting the timer. The timer will be reset when there is a read followed by a write of the Rail Voltage Watchdog Register (RVWR) in the FPGA. The uC has direct turn-off control of the rail voltage by writing a zero to bit 2 in the FPGA control register. This bit is ANDed with the rail voltage watchdog timer to enable the rails.

Power amplifier output currents should be read and verified by the uC at least at the following times:
1. A special diagnostics routine should sequentially drive all coil output channels to 80% full scale and verify the output current (testing the channels sequentially so as not to exceed the 12 amp limit).
2. If the uC reads a heatsink temperature exceeding OT2 then the uC should read all coil currents and voltages, remove power from the coil drive amplifiers (rails), and log the fault in the message log. 3. Periodical verification of correct coil currents.

Heatsink temperature should be read by the uC within every 2 seconds.

After a reset, the power supply will deliver the last programmed current values to the coils, when the supply is used with an LX system.

RTD MONITOR

An 8 channel temperature monitor is included to eliminate the cost of a Fluke temperature monitor presently installed on HFO systems. The board will interface to 4 wire 100 ohm platinum RTD's at the magnet. The accuracy over a +/- 20C board temperature range, relative to the board temperature when calibration was performed, is +/-0.7C. This accuracy is for a 4 wire RTD connected to the rear of the chassis, not at the magnet through penetration wall filters, the error from the filters and interference from the MRI system are not well defined. The normal magnet operating temperature range is within 15C – 45C; temperature measurements outside this range should be logged as a sensor fault (i.e. open sensor, shorted sensor). A 16 bit ADC will be used to acquire temperature data with a temperature resolution finer than 0.01 degree centigrade. The uC will monitor the board temperature from a sensor located near the RTD monitoring circuitry, within every 2 seconds. This information is used to compensate the RTD temperature measurements for changes in chassis temperature.

GATING

Data in window (DIW), and Z2 Enable signals enter the board via connector J8. The DIW and Z2 enable signals are currently not used. The original intent of the DIW signals was to prevent shim current updates during the MRI data acquisition time. J8 also contains 2 differential signals, which can be programmed to be either inputs or outputs.

POWER AMPLIFIER INTERFACE

DACS AND POWER AMPLIFIERS

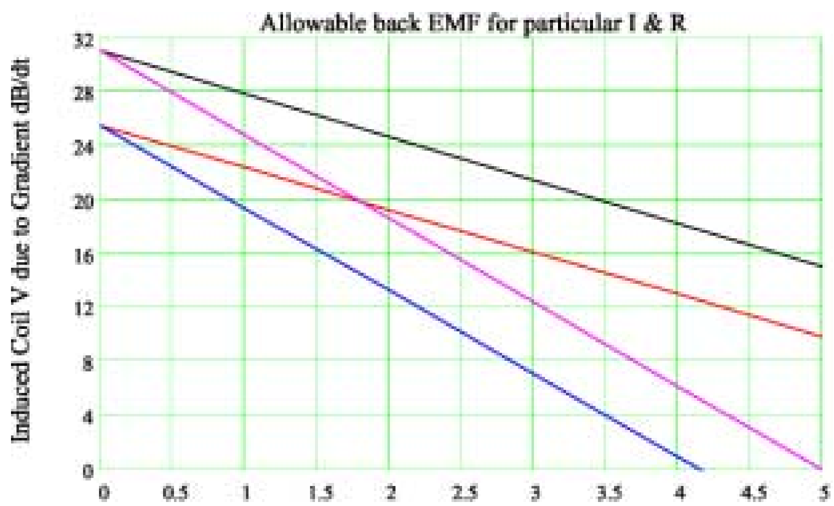
The board contains 6, offset binary, 16 bit DAC's. The FPGA re-formats the twos compliment data, from the uC, into offset binary data before sending it to the DAC's. There is 1 DAC for each coil drive output. The uC writes DAC values to registers in the FPGA (See FPGA memory map section). After the uC has completed the write cycle to an FPGA DACn register, the contents of all the FPGA DACn registers will be transferred to a shift register and shifted out to all the DAC's, serially at a 750KHz bit rate. The output of all the DAC's will be updated within 200uS of an FPGA DACn register update by the uC. The output currents will not settle for up to 20mS (5*L/R for 20mH, 5ohm coil). If the Gate bit (CREG[9]) is set, the DACs will not get loaded while the Data In Window (DIW) is asserted.

Each output has a single power op-amp to drive +/-4 amps coil current. The gain is set to achieve +/-5 ampere at full scale of the DAC. This extra gain is to allow compensation for DC offsets and gain errors. The power amplifiers, along with associated circuitry, are powered from a supply, which is isolated from ground. All communication between the digital and analog electronics is through optical devices. The power amplifiers are protected from high voltage transients on their outputs by rail clamp diodes. The coil drive electronics are protected from a short to ground. A short to ground will cause little current flow due to the isolated power supply for the coil drive electronics.

There is a minimum load inductance that must be present on each load to prevent the outputs from oscillating. Channel 1 must have at least 3mH and the other channels must have at least 0.5mH. The uC must monitor for oscillating channels and turn off the rails if an oscillation occurs. The circuit board can accept 2 different types of power amplifiers, an OPA541 and a PA39. The PA39 has a feature where the front end amplifier (via the +/-Vb pins) can be powered from the same voltage as the output transistors, or from a higher voltage. When connected to the same voltage, the output swing is limited to approximately 5.5 volts below the rail voltage. When +/-Vb is 10 volts greater than the rails, the output transistors can be driven into saturation, allowing the output swing to go to the rail voltage. The following chart shows the constant current mode, operating region of the supply channels. This region is determined by the amplifier load resistance, coil current, and induced coil voltage due to gradient dB/dt. Notice that when Vb is 10 volts above Vs, we can tolerate about 5.5 volts more induced voltage.

For the supply to remain in constant current control the induced voltage from the shim coil, due to gradient dB/dt, must fall below the curves in the following chart.

Operating area is beneath the curve



Coil Current supplied by shim supply

R=3; Vb=Vs R=6; Vb=Vs R=3; Vb=Vs+10 R=6; Vb=Vs+10

Coil Current Monitoring

Current monitoring is achieved by measuring the voltage across each current sense resistor. This is done by buffering the current sense signals, and multiplexing (ADG408) them to a high impedance amplifier which adds gain and filtering before driving the input of a 16 bit ADC. The filter is a 2 pole Butter-worth with a cutoff frequency of 1700Hz. This filter reduces not only the random background noise to the ADC input, but also attenuates the 63KHz noise generated by the ACCD amplifiers by about 63dB. Six of the AD408 inputs will be signals from the current sense resistors, and the remaining 2 inputs will be AGND and a reference voltage. The reference will be ½ the ADC reference value. The ground and reference voltage values will be used to remove offset and gain errors from the mux and amplifier. Selecting accurate sense resistors (<0.5% tolerance), instrumentation amplifiers, and ADC voltage reference, will allow errors in the power electronics and DAC's to be removed electronically. After a reset, or when a DAC calibration command is received by the uC, calibration data should be collected. There are 2 sets of calibration data that must be collected, 1 set is offset and gain data for the current monitoring side and the other set is for the current drive circuit.

The following **Current Monitoring Calibration** data must be collected:

1. Select the AGND input of the DG408 and sample the level. This is the **Current Monitors DC Offset (CMDCO)** of the DG408, amplifier, and the ADC. This value will be subtracted from all other future samples.
2. Select the Reference input of the DG408 and sample the level. After removing the DC offset (see 1 above) the digital value should be 16384d. Find the **Current Monitors Gain Error Ratio (CMGER)** by dividing 16384 by the actual value measured. This gain error ratio will be multiplied by all future samples (after the offset has been removed).

The following **Current Drive Calibration** data must be collected:

1. Set all the DAC outputs to zero. For each shim coil drive circuit, select the shim coil input of the DG408 and sample the level. Correct for current monitor errors by subtracting the CMDCO and multiplying the result by GMGER. The resulting value is the DC offset of the power electronics. **This is the Power Electronics DC Offset (PEDCO)** value and must be subtracted from the values being sent to the DAC for that channel. Repeat for all channels.

2. For 1 channel at a time, set the DAC value to ½ full scale (2.5 amperes; 16384d) after subtracting off the PEDCO. Sample the level and correct current monitor errors by subtracting the CMDCO and multiplying by CMGER. Find the **Power Electronics Gain Error Ratio (PEGER)** by dividing 16384 by the actual value measured. This gain error ratio will be multiplied by all uncompensated DAC values (after the PEDCO has been removed) to yield the compensated value. During normal operation, the compensated values are sent to the DAC.

The FPGA reads both the current ADC and voltage ADC values at the same time. Data from the two, 8 channel, 16 bit ADC's is shifted through a single serial interface into the FPGA at a 750KHz bit rate. It takes about 352uS to acquire and get the 16 values of coil current and voltage data into the FPGA registers. The settling time of the filter, being on the ADC side of the MUX, must be included each time a new input is switched to the ADC via the MUX. The filter settling time for the current monitoring filter is about 5/fc = 5/1700Hz = 2.94mS (5 time constants results in a 0.8% settling error).

Example:

0.352mS (time to do 8 sets acquisitions and read all 16 bits per acquisition) 26.000mS (Settling time for 8 signals multiplexed at 2.94mS per switch; not including the coil current settling time, which can approach 20mS, after a DAC update) 26.352mS Total

The coil current and voltage will be sampled and made available to the uC every 43.69mS (2^19/12MHz).

UC SYNCHRONIZATION

Either of the following methods can be used to synchronize data updates with uC reads:

1. Polling mode
- Each time the uC reads one of the coil current or voltage registers the IDR flag (bit 5 in the FPGA status register) is reset.
- After new data is loaded in the current and voltage registers, in the FPGA, this bit will be set.
- The uC can poll this bit after a read to determine when new data is available.

2. Interrupt mode.
- If interrupt enable bit IIE (bit 5 in the FPGA control register) is set then the FPGA will generate an uC interrupt after new data is loaded into the coil current and voltage registers.
- The uC responding to the interrupt will read the FPGA status register to determine the nature of the interrupt. If the IDR (bit 5 of the status register) is set then the uC will read the current and voltage data, and write a 1 to IACK bit (bit 6 in the FPGA control register).

COIL VOLTAGE MONITORING

Coil voltage monitoring is achieved by measuring the voltage across each coil. This is done with instrumentation amplifiers connected across each current sense resistor, via an attenuator. The attenuated coil voltage signals are multiplexed (ADG408) and sent to a high impedance amplifier, which buffers and filters it before driving the input of the 16 bit ADC. The filter is a 2 pole Butter-worth with a cutoff frequency of 1100Hz, and provides 71dB of attenuation at 63KHz. Six of the AD408 inputs will be signals from the coil voltage, and the remaining 2 inputs will be attenuated supply voltages to the power op-amps. A 5% accuracy is good enough for these measurements so no calibration is required. As with coil current monitoring, the settling time of the filter must be included each time a new MUX input is switched to the ADC. The filter settling time for the voltage monitoring filter is about 5/fc = 5/1100Hz = 4.55mS (for a 0.8% settling error). The settling time for all 8 channels is 36.4mS. Adding the 352uS acquire and shift time (from the current monitoring section) gives 36.752mS, which is less than the 43.69mS update rate for the current and voltage registers, indicated in the current monitoring section.

POWER AMPLIFIER SPECIFICATIONS

Parameter	Test Condition	MIN	TYP	MAX	UNIT
Output Current (Each Channel)		0		+/-4	Amps
Cumulative Current	Summed absolute value current from all Channels	0		12	Amps

Output Voltage (Each Channel)		0		+/-30	Volts
Full PWR Bandwidth of error signal path		20			KHz
Duty Cycle		100			%
DC Offset	% of full scale			+/-.3	%
Gain Error	% of full scale			+/-1	%
Output drift	% of full scale, over 24 hours at room temperatur			200 (.8mA)	PPM
Output Noise (Note 2)	Peak-peak full scale, 0.1 to 10Hz			15	PPM
	Peak-peak full scale, 10 to 50KHz			50	PPM
	Above 20MHz (Note 1)			-90	dBm
Load Resistance	Must be no load stable	5.0		6.0	Ohms
Load Inductance	Channel 1	3.0		20	mH
	Channels 2 through 6	0.5		20	mH
Load Capacitance	To ground	0.0		0.1	uF
Rise time (10-90%)	0 to 4 amps @ worse case load			8	mS
Settling time	From 0 to 15 PPM of Full			35	mS
Resolution	DAC step			0.15	mA
DAC update delay	From time uC writes to DAC registers in FPGA		200		uS
I/V sample rate (all channels)	Time between updates available to uC		43.7		mS

Note 1:

Assumes:

1. 160 dBm power into preamp to generate artifact in image.
2. 40 dB penetration wall filtering 30 dB coupling loss between coil and unshielded shim circuit conductors. This coupling loss would be much greater if the shim cable shield was connected to the RF shield, and the coil was behind the shield (facing away from the RF coil). Unfortunately the current practice is for the cable shield to stop short of extended RF shield.

3. 4uA/PPM using a full scale current of 4 amps.

RTD INTERFACE

The board contains 8 inputs for 4 wire, platinum, 100 ohm, RTD sensors having a 0.00385 ohms per ohm degree temperature coefficient. There are individual 2 mA current sources for each channel. Signals are received and gained 10x by LT1167 instrumentation amplifiers. The LT297 contains 2 amplifiers. One amplifier adds gain and removes the zero degree C offset while the other inverts and filters the signal. The filter is a 1 pole Butter-worth filter, with a cutoff frequency of 2300Hz, providing ~31dB of attenuation at 63KHz. The output of the filter provides an ADC input voltage of zero to 10 volts which corresponds to an RTD temperature range of approximately 0 - 62C. The ADC is a 16 bit LT1609 connected for unipolar operation. This results in a temperature resolution of approximately 0.001 degrees per ADC step. The current source and the LT1167 are the major contributors to drift error. This error is less than +/-1.0C over a +/-20C change in circuit temperature, relative to the temperature when the calibration data was collected. Software improves on this accuracy by correcting the measured temperature for circuit board temperature changes.

A provision has been included for installing a cover over the RTD monitor circuitry to shield it from abrupt changes in temperature due to cooling fan draft.

The FPGA reads the ADC values from all 8 RTD channels sequentially. Data from the 16 bit ADC is shifted through a serial interface into the FPGA at a 750KHz bit rate. It takes about 176uS to acquire and get the 8 values of data to the FPGA registers, not including the filter settling time. The settling time of the filter, being on the ADC side of the MUX, must be included each time a new MUX input is switched to the ADC.

Example:

0.176mS (time to do 16 acquisitions and read 16 bits per acquisition)
16.667mS (Settling time for 8 signals multiplexed at 2.1mS per switch 8*5/.0021) 26.352mS Total

The RTD values will be sampled and made available to the uC every 43.69mS (2^19/12MHz).

RTD COMPENSATION

General

This is done at the board house. The 2 main errors in the circuit are gain and offset errors. These errors can be characterized and corrected digitally by the uC. The following steps are required for each RTD channel:

1. Acquire temperature data at 2 different RTD temperatures (Use 0.02% precision resistors to

   - 105.00 ohms => 12.987 degrees C
   - 118.00 ohms => 46.753 degrees C

2. Store the data in EEPROM along with the board temperature data. Using the 2 points saved in EEPROM, linear interpolate real RTD data to remove the gain and offset errors.

There are eight, 16 bit registers in the FPGA (RTD[8:1]) which hold straight binary RTD data.

Additionally, board temperature related errors are reduced by applying the following correction to the measured temperature values before they are sent to the host.

Reported_Temp = Corrected_RTD_Temp + (19.10E-3 * (Stored_BtempR - Present_BtempR))

Where:
Reported_Temp: Is RTD temperature sent to the Host/MGD.
Corrected_RTD_Temp: Is the RTD temperature after gain and offset errors are removed. Stored_BtempR: Is the board temperature stored in nonvolatile memory.
Present_BtempR: Is the current value of the board temperature.

Present_BtempR should be updated at least every 30 seconds.

UC SYNCHRONIZATION

Either of the following methods can be used to synchronize data updates with uC reads:

1. Polling mode
   - Each time the uC reads one of the RTD registers the RDR flag (bit 4 in the FPGA status register) is reset.
   - After new data is loaded in the FPGA's RTD registers this bit will be set.
   - The uC can poll this bit to determine when new data is available.
2. Interrupt mode
   - If interrupt enable bit RIE (bit 4 in the FPGA control register) is set then the FPGA will generate an uC interrupt after new data is loaded into the RTD registers.
   - The uC responding to the interrupt will read the FPGA status register to determine the nature of the interrupt. If the RDR (bit 4 of the status register) is set then the uC will read the RTD data, and write a 1 to IACK bit (bit 6 in the FPGA control register).

Open and shorted sensors will cause the ADC counts to be nearly full scale or zero respectively. Software should assume a sensor fault condition when the measured temperature is out of the 15-45 degree range. For example, report an open sensor if the RTD temperature exceeds 50C and a shorted sensor if it temperature is less than 10C. It should be noted that non-sensor faults could give similar symptoms as a defective sensor. For example, an open in the sense wire can look like an open or shorted sensor, depending on where the input amplifier floats.

RTD MONITOR SPECIFICATIONS

Parameter	Test Condition	MIN	TYP	MAX	UNIT
Number of channels		8			

temperature monitoring range		10		50	Degree C
Upper sensor fault threshold	Above which sensor fault is assumed and logged		45		Degree C
Lower sensor fault threshold	Below which sensor fault is assumed and logged		15		Degree C
Accuracy (note 2)	Within board temperature range of 5-45 degrees C	+/ 0.7			Degree C
ADC full scale temperature				62	Degree C
Sample period	Per channel			0.75	S/chan
ADC/filter settling time	Time from changing RTD input via MUX and stable value at ADC		2.2		mS
Noise (output)	Above 20MHz (Note 1)			-87	dBm
RTD sample rate (all channels)	Time between updates available to uC		43.7		mS

Note 1

Assumes:

1. -160 dBm power into preamp to generate artifact in image.
2. 23 dB penetration wall filtering
3. 20 dB coupling loss between coil and unshielded shim circuit conductors.
4. >30dB RF shield effectiveness

Note 2

The stated accuracy includes compensating for board temperature.

GATING

Data in window (DIW), and Z2 Enable signals enter the board via connector J8. The Z2 enable signals are currently not used. The original intent of the DIW signals was to prevent shim current updates during the MRI data acquisition time. J8 also contains 2 spare differential signals, which can be programmed to be either inputs or outputs. If the Gate bit (bit 7 in the FPGA control register) is set then DACs will not be updated while the DIW flag (bit 1 in the FPGA status register) is set. If the gate bit is reset, then the DIW signal has no effect. Due to the long time constant of the large shim coil inductance, the DIW signal will probably never be used for this purpose.

UC SYNCHRONIZATION

Either of the following methods can be used to synchronize data updates with uC reads:

1. Polling mode
   - The DIW flag is set during data acquisition time of the MRI scanner.
   - This bit can be polled by the uC.
2. Interrupt mode
   - If interrupt enable bit DIE (bit 3 in the FPGA control register) is set then the FPGA will generate an uC interrupt when the DIW flag is set.
   - The uC responding to the interrupt will read the FPGA status register to determine the nature of the interrupt. If the DIW flag (bit 3 of the status register) is set then … The DIW is not a register output, and therefore not cleared by setting the IACK bit.

SYSTEM MONITORING

The uC is equipped with 16, 10 bit ADC inputs. 6 of these channels are connected as follows.

ADC Channel	Description
0	½ VCC (2.5V nominal)
1	Output of LM35 board temperature sensor near Power amps. Vo = 10mV/ degree C
2	Output of LM35 board temperature sensor near RTD circuit. Vo = 10mV/ degree C
3	Output of AD22103KT temperature sensor on air inlet side of heatsink. Vo = 28mV/C + 0.25V
4	Output of AD22103KT temperature sensor on air outlet side of heatsink. Vo = 28mV/C + 0.25V
5	Ground
6	2.5 volt supply

Heat sink and temperature is measured with AD22103KT temperature sensors. J-18 is located near the air outlet end of the heatsink. J-17 is located near the air inlet of the heatsink. Comparing the temperatures of the probes may provide an in**3.7. FPGA**

The board has an Altera EP1K100_208-3 field programmable gate array on board. Programming is done from the EPC2 after every reset or power cycle. The EPC2 is programmed by a byteblaster connected to header J19 or from the uC via an IO address. An analog switch in front of the EPC2, controlled by the state of a control register bit in the PLD, determines which device can program the EPC2. The FPGA controls all the communication to/from the ADC's and DAC's.

JTAG INTERFACE

(Joint Test Action Group developed a standard for boundary-scan testing (BST) of printed circuit board).

There are 4 JTAG compatible devices on the board. The uC will have it's own standard JTAG interface with a 14-pin connector (equivalent to an AMP 103149-7) that conforms to the IEEE Standard 1149.1. The JTAG port will not be accessible from outside the module; therefore, it will only be used for downloading the basic core software program during manufacturing, and for bench testing. A second header will be present for the EPLD, FPGA, and the EPC2, and will primarily be used for in circuit programming of these devices.

SYSTEM RESET

There are 5 reset sources for the board:

1. Push button reset switch (via supervisor)
2. 3.3 volt supply too low (via supervisor)
3. Host RS-232 break detection circuit
4. Internal uC watchdog timeout.
5. All sources are active low. Signals from sources 1 through 4 are ANDed in the PLD to generate the FPGA_RESET_N and RESET_DSP_N signals. RESET_DSP_N is a bi-directional signal, which can be driven by the uC or the PLD (See Internal Watchdog section for additional information). The uC asserting this signal will also assert FPGA_RESET_N.

HOST RS-232 BREAK DETECT

The break detect reset will be generated if there is a high on the RS-232 receive signal for more than approximately 6mS.

I/O PIN ON UC

The I/O pin reset is an output of an edge triggered one shot inside the PLD, triggered by a low transition of the I/O pin (IO_RESET signal on pin 56, IOPA6) on the uC.

INTERNAL UC WATCHDOG TIME OUT

The reset signal to the uC is pulled up to 3.3 volts by a resistor, and the PLD output driving this signal is programmed to be an Open Collector output. A uC watchdog timeout will cause the uC to assert the active low reset signal while the uC resets itself. The FPGA reset (FPGA_RESET_N) will be asserted whenever the RESET_DSP_N signal is asserted.

FPGA PROGRAM

The FPGA is downloaded automatically by the EPC-2 (programmed at the board house or later via the JTAG port) after a board reset. Boot of the uC and programming of the FPGA is done at the same time. The uC can monitor the state of the FPGA's configuration signals via a register in the EPLD. If the uC is to program the FPGA it will set the FCS (FPGA_CONFIG_SELECT) bit, in the EPLD control register, and read/write the FPGA configuration signals via registers in the EPLD. Setting the FCS bit will enable the EPLD to drive the appropriate configuration signals, and set the FPGA_CONFIG_SELECT signal to disconnect the EPC2. The FCS bit defaults to the EPC2 mode after a reset. See the memory map section, and Altera literature for additional information on the FPGA download details. For cost savings the EPC2 device may be left off the board on production systems.

RS-232 SERIAL PORTS

There are 2 RS-232 ports on the board. The port at J1 is for connection to the Host computer, and the port at J2 interfaces to the SCI pins on the uC and is defined as the debug port. Both ports have similar baud rates, but do not have the same pinout.

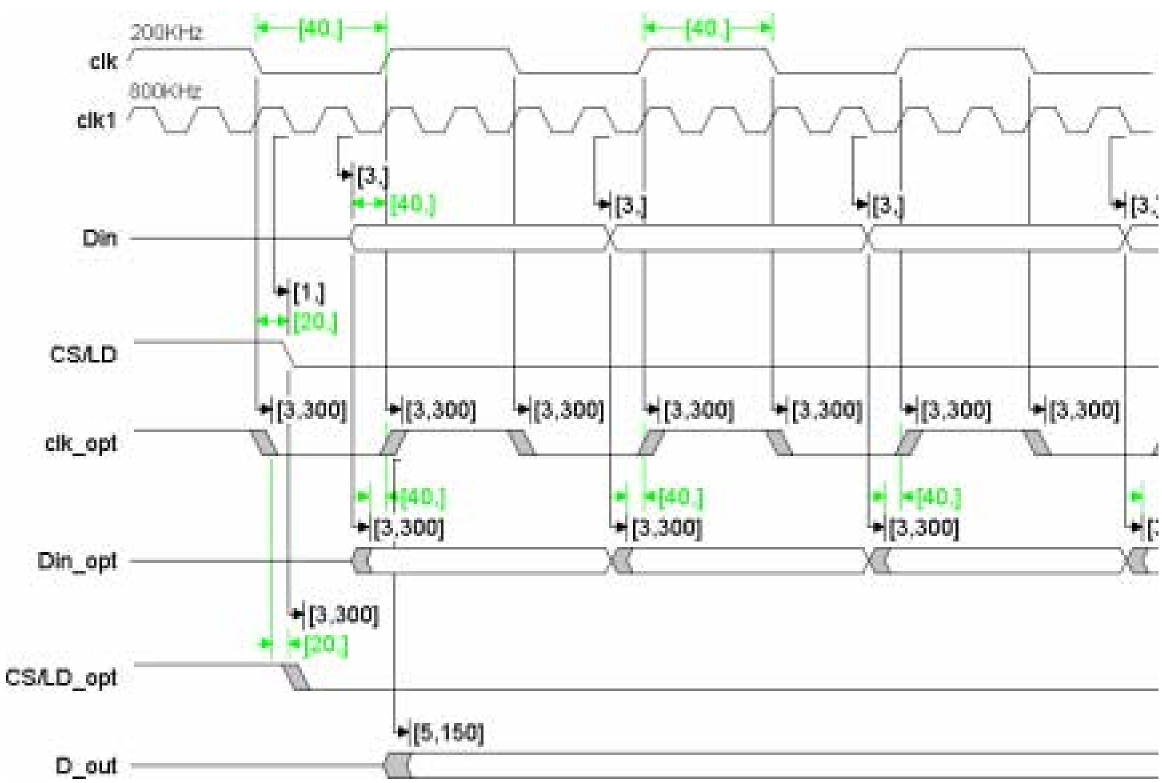
- Baud rate 19,200

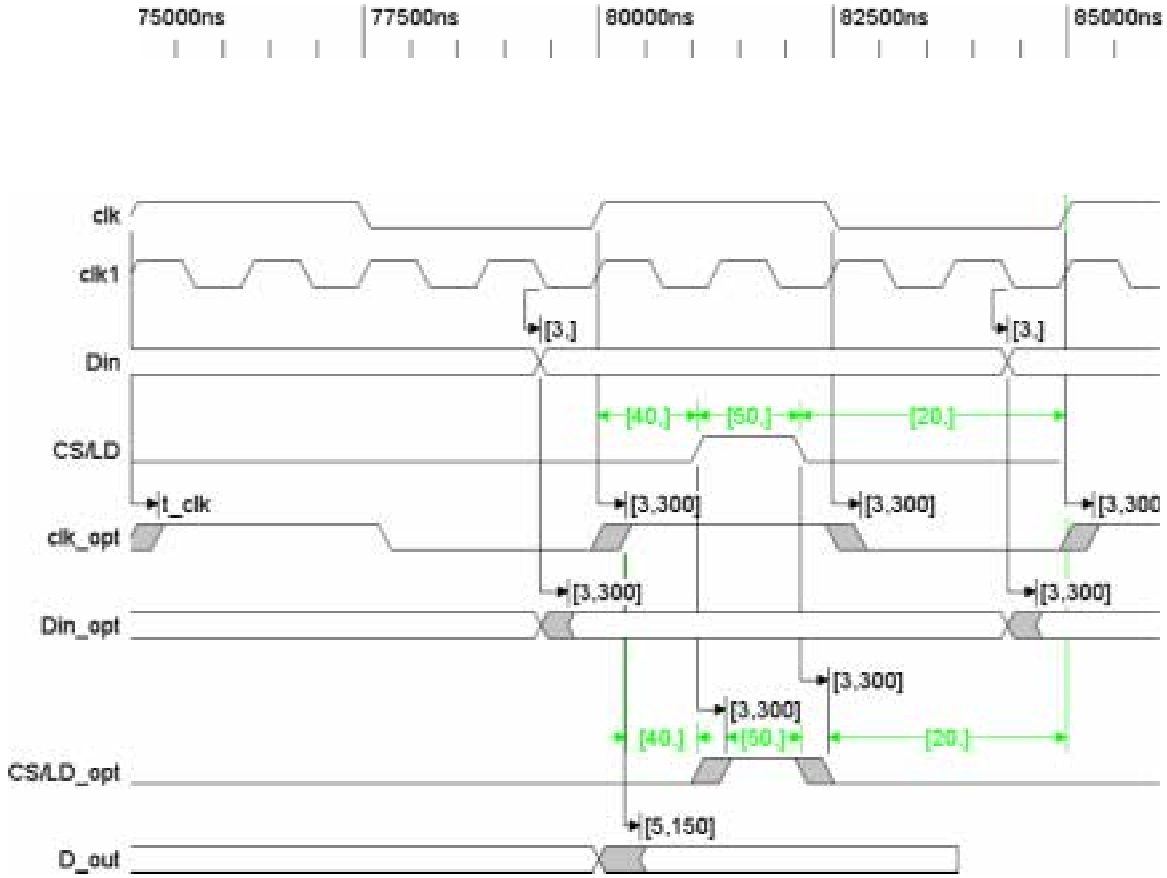
- Date bits 8
- Stop bits 1
- Start bits 2
- Parity None
- Control Implemented in

CAN PORT
The Shim Supply will use CAN for all its communications with the Excite system. CAN is a fast and reliable medium for communication between multiple systems. CANopen is the specific protocol that is used at MR.
See the document *CANopen Implementation for MR* for more details. The *Communication Core SDD* document contains specific information about the setup and operation of the MR CAN network.

4. SYSTEM TIMING 4.1. DAC INTERFACE

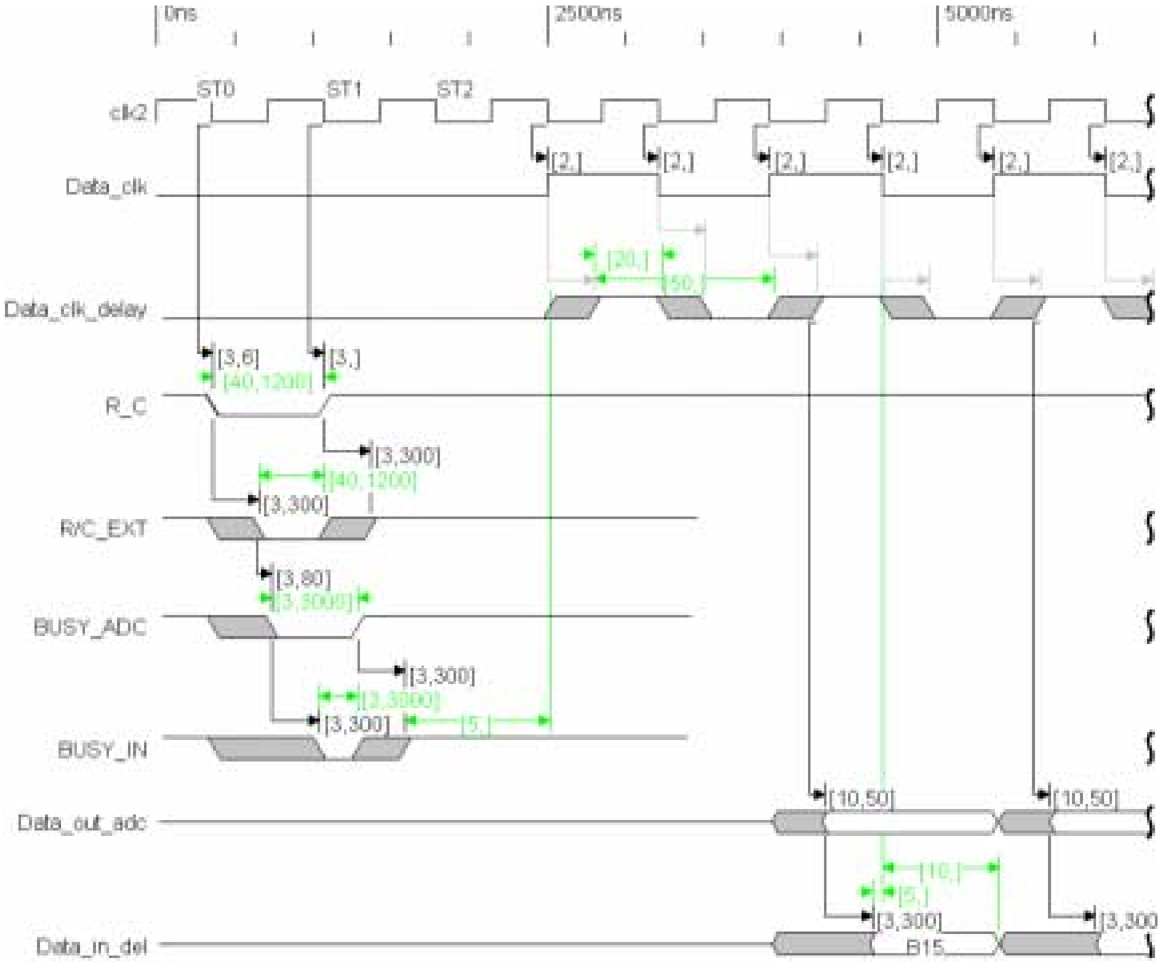
15us





75000ns
77500ns
80000ns
82500ns
85000ns
clk
clk1
[3,]
[3,]
Din
[40,]
[50,]
[20,]
CS/LD
t_clk
[3,300]
[3,300]
[3,300
clk_opt
[3,300]
[3,300]
Din_opt
[3,300]
[3,300]
[40,]
[50,]
[20,]
CS/LD_opt
[5,150]
D_out

ADC INTERFACE



ELECTRICAL INTERFACES

J#	Connector Type	Description
J1	DB9M	Host RS-232 interface
J2	DB9M	Debug/monitor RS-232 interface
J3	37 pin circular	Amp Circular – Cabled to PWA
J4	DB9F	CAN
J5	DB9M	CAN
J6	DB37F	RTD interface
J7	DB9F	Z2 (For HFO compatibility) – Cabled
J8	DB9M	Enables – connected to PWA via ribbon cable

In the following tables, input (I) specifies a signal that is coming into the board from a peripheral. An output (O) is a signal originating on the board and going to a peripheral. A pin not used on the connector is specified as NC (Not Connected).

J1- HOST RS-232 INTERFACE (COM1)

This through hole connector is top of a stacked (shared with J2) 9 pin male right angle board mount sub-D connector with board locks and threaded inserts. The pinout follows:

Pin	Name	Description
1	DCD	Not used
2	RX	RS-232 RX from host TX
3	TX	RS-232 TX to host RX
4	DTR	Wired to pin 6
5	GND	RS-232 GRD
6	DSR	Wired to pin 4
7	NC	-
8	NC	-
9	NC	-

J2- DEBUG/MONITOR INTERFACE (COM2)

This through hole connector is the bottom of a stacked (shared with J1) 9 pin male right angle board mount sub-D connector with board locks and threaded inserts. The pinout follows:

Pin	Name	Description
1	NC	-
2	TX	RS-232 TX
3	RX	RS-232 RX
4	NC	-
5	GND	RS-232 GND
6	NC	-
7	NC	-
8	NC	-
9	NC	-

J3- SHIM COIL INTERFACE CONNECTOR

This is a female 37 pin through hole plastic circular connector (AMP 206306-1). An internal wiring harness connects Mate-N-Lok connectors on the board (1 per channel) to this circular connector. The pinout follows:

Pin	Board connector & Pin number	Name	Description
5	J10-2	CH1	Z2
6	J10-1	CH1 RTN	Z2 return
7	J11-2	CH2	Z3
8	J11-1	CH2 RTN	Z3 return
17	J12-2	CH3	ZX
18	J12-1	CH3 RTN	ZX return
19	J13-2	CH4	ZY
20	J13-1	CH4 RTN	ZY return
25	J14-2	CH5	X2-Y2
26	J14-1	CH5 RTN	X2-Y2 return
27	J16-2	CH6	XY
28	J16-1	CH6 RTN	XY RTN
37	37	CGND	Chassis Ground
Other pins	Other pins	NC	

J4- CAN

This is the top of a stacked connector shared with J5. This through hole connector is a female 9 pin right angle sub-D type connector with board locks and threaded inserts. The pinout follows:

Pin	Name	Description
1	NC	
2	CANL	CAN high side of differential signal
3	CPWR RTN	Return for isolated 24volt signal
4	NC	
5	CPWR RTN	Return for isolated 24volt signal
6	CPWR RTN	Return for isolated 24volt signal
7	CANH	CAN high side of differential signal
8	NC	
9	CPWR	Isolated 24 volts from off board

This is the bottom of a stacked connector shared with J4. This through hole connector is a male 9 pin right angle sub-D type connector with board locks and threaded inserts. The pinout follows:

Pin	**Name**	**Description**
1	NC	
2	CANL	CAN high side of differential signal
3	CPWR_RTN	Return for isolated 24volt signal
4	NC	-
5	CPWR RTN	Return for isolated 24volt signal
6	CPWR_RTN	Return for isolated 24volt signal
7	CANH	CAN high side of differential signal
8	NC	-
9	CPWR	Isolated 24 volts from off board

J6- RTD INTERFACE

This through hole connector is a standard 37 pin female right angle board mount sub-D connector with board locks and threaded inserts (AMP 747847-4 or eq.). The pinout follows:

Pin	Name	Description
1	SENSE_1-P	RTD Channel 1
20	SENSE_1-N	
2	SOURCE_1-P	
21	SOURCE_1-N	
3	SENSE_2-P	RTD Channel 2
22	SENSE_2-N	
4	SOURCE_2-P	
23	SOURCE_2-N	
5	SENSE_3-P	RTD Channel 3
24	SENSE_3-N	
6	SOURCE_3-P	
25	SOURCE_3-N	
7	SENSE_4-P	RTD Channel 4
26	SENSE_4-N	
8	SOURCE_4-P	
27	SOURCE_4-N	
9	SENSE_5-P	RTD Channel 5
28	SENSE_5-N	
10	SOURCE_5-P	
29	SOURCE_5-N	
11	SENSE_6-P	RTD Channel 6
30	SENSE_6-N	
12	SOURCE_6-P	
31	SOURCE_6-N	
13	SENSE_7-P	RTD Channel 7

32	SENSE 7-N	
14	SOURCE 7-P	
33	SOURCE 7-N	
15	SENSE 8-P	RTD Channel 8
34	SENSE 8-N	
16	SOURCE 8-P	
35	SOURCE 8-N	
17	NC	-
36	NC	-
18	NC	-
37	NC	-
19	NC	-

J7- Z2 INTERFACE FOR HFO COMPATIBILITY

This connector is a female 9 pin chassis mount sub-D type connector with threaded inserts. An internal wiring harness connects a Mate-N-Lok connector on the board to this sub-D connector. The pinout follows:

Pin	Board connector & Pin number	Name	Description
1	J15-2	CH5	Z2, same as J3-25
2	J15-2	CH5	Z2
3	J15-2	CH5	Z2
4	4	NC	-
5	5	NC	-
6	J15-1	CH5 RTN	Z2 RETURN, same as J3-26
7	J15-1	CH5 RTN	Z2 RETURN
8	J15-1	CH5 RTN	Z2 RETURN
9	9	NC	-

J8- ENABLES

This connector is a male 9 pin chassis mount sub-D type connector with board locks and threaded inserts. Header J5 on the board supplies these signals. The pinout follows:

Pin	Name	Description
1 (1)	Spare-IO1-P	Spare – active high part of
6 (2)	Spare-IO1-N	Spare – active high part of
2	NC	Was shim on signal to enable NAV shims
7 (4)	GND	Ground reference for DIW

3 (5)	DIW	TTL active high Data in window signal from IPG, not used
4 (7)	Spare-IO2-P	Spare – active high part of
9 (8)	Spare-IO2-N	Spare – active high part of
Other pins	NC	-

Note: the pin number in () is the pin number for the header.

JTAG FOR UC 14 pin header

Pin	Signal Name	Description
1	TMS	I
2	/TRST	I
3	TDI	I
4	GND	Power
5	+3.3VDC	Power
6	NC	Clipped Pin
7	TDO	O
8	GND	Power
9	TCK_RET	O
10	GND	Power
11	TCK	I
12	GND	Power
13	EMU0	I/O
14	EMU1	I/O

Table 7 – uC JTAG Interface

The +3.3VDC and Ground will be supplied from the DSP board to the JTAG interface.

JTAG FOR PLD, FPGA & EPC2 14 pin header

Pin	Signal Name	Description
1	TCK	I
2	GND	Power
3	TDO	O
4	+3.3VDC	Power
5	TMS	I
6	+3.3VDC	Power
7	NC	Clipped Pin
8	NC	-
9	TDI	I
10	GND	Power
11	NC	Clipped Pin
12	NC	Clipped Pin
13	TRST	I
14	NC	-

Table 8 – uC JTAG Interface

JUMPERS
Jumpers on the analog output section of the board are for development only and will be removed and shunted by 0 ohm resistors for product. J5 is installed only when programming the PLD and FPGA.

LIGHT EMITTING DIODES (LEDS)

There shall be 9 LED's on the front of the chassis. The description of these LED's follow:

Position from front left	Signal	PWB Designator	Description
1	Heartbeat	DS2	CPU pulses periodically
2	Comm Error	DS3	On if CAN or other Error – from
3	Comm Stat	DS4	CAN or other Status – from
4	Power OK	DS5	On if +/-31V power supplies are
5	FUNC ERR	DS7	Undefined - from uC
6	OUT EN	DS8	On when power to coil drive amps is
7	PLD Watchdog	DS9	On if PLD watch dog timer is running
8	SEQ ON	DS10	Blink indicates FPGA reading and
9	5V	DS11	On indicates +5V supply is

Table 8 – Front Panel LEDs

There are 2 LED's located near the CAN interface electronics which are not viewable without opening the enclosure. These are:

Signal	Designator	Description
CAN TX	DS1	CAN transmit LED

CAN RX	DS6	CAN receive LED

SWITCHES

There shall be one quad DIP switch that penetrates the front of the chassis. These switches are connected between ground and pull-up resistors connected to the EPLD. Note that the switch state “Closed” (Down) equates to a 0 and “Open” (Up) to a 1.

Switch	Description
1	Not used
2	CAN node select address; Open = 16H, Closed = 17H
3	Lx/MGD; Open = MGD, Closed = Lx
4	Not used

A reset switch will be placed between ground and the reset in pin on the supervisory circuit. This switch will not be accessible from outside of the module.

TEST POINTS

The following test points are on the board:

Signal Name	Description
+31V	Power
-31V	Power
+12V	Power
-12V	Power
+5V	Power
+3.3V	Power
GND	Power
CH1	COIL
CH1 RTN	COIL
CH2	COIL
CH2_RTN	COIL
CH3	COIL
CH3 RTN	COIL
CH4	COIL
CH4_RTN	COIL
CH5	COIL
CH5 RTN	COIL
CH6	COIL
CH6_RTN	COIL

Table 7 – Test Points

ENVIRONMENTAL REQUIREMENTS

The shim power supply must operate in the environmental conditions specified below. The sub-system will be tested in an environmental chamber.

OPERATING CONDITIONS

Ambient Temperature:

5 deg C to 40 deg C Ambient
Temperature Stability: ~ 3
deg C/hr
Humidity (non-condensing): 10 to 90 %RH
Magnetic Field: ~ 50 Gauss
Altitude: -230 m to 3000 m

OVER CURRENT PROTECTION

The AC mains are protected by fuses in the power input module. If the fuses need to be replaced, they must be replaced a with slow blow type of at least 6.2 amps. Type MDL8A, 8 amp fuses are acceptable.

INSTALLATION AND SERVICE REQUIREMENTS

It is intended that the shim power supply be a FRU, The average replacement time for the assembly should be less than 15 minutes. No calibration will be required for the electronics on this assembly. A setup to the assembly will require a label stating that setup is required and what the setup is and the choices.

The Mean Time Between Failures (MTBF) goal for the assembly is > 20,000 hours of use.

- Fan: 60,000 hours
- Power Supply: 200,000 hours

The fan is a FRU and will be mounted to its own, easily removable bracket, for quick field replacement.

All switches, LED's, jumpers, and test points will be available outside the assembly panel with descriptive labeling so the assembly does not need to be opened.

The assembly will provide diagnostic capabilities that allow isolation of failure to 70% accuracy to the FRU assembly level.

Software must be available to allow a service engineer to monitor the shim supply voltages, temperatures, and latest faults, via a PC connected to the serial debug port. Additionally, this information must be available via the C shell and the host interface.

BOARD INPUT POWER POWER

ESTIMATE

The system will require the following power and current:

Voltage (V)	Maximum Power (W)	Maximum Current (A)
+5 VDC	3.0	0.6
+15 VDC	6.0	0.4
-15 VDC	3.0	0.2
+31 VDC	388	12.5
-31 VDC		

Maximum Power

POWER SUPPLY SPECIFICATIONS

The chassis will accept both the MP6 chassis from Astec and the Vega chassis from Lambda. The following table lists the specifications for both supplies.

Specification	Astec MP6	Vega
Voltage	85-264 VAC	85-264 VAC
Frequency	47-440 Hz	47-63Hz
Inrush Current	40A max	< 40 A cold start
Fuse	15A Internal	Internal, no access
Efficiency	70% min @ 115VAC	75% typical @ 230VAC
Power factor	0.99 typical	